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University spin-out companies and venture capital

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Abstract

The creation of university spin-out companies that create wealth is a major policy objective of governments and universities. Finance is a catalyst of this wealth creation yet access to venture capital is a major impediment faced by these companies. In this article we adopt a finance pecking order perspective to examine the problems faced by those university spin-out companies seeking to access venture capital. We triangulate evidence from spin-out companies, university technology transfer offices and venture capital firms in the UK and Continental Europe to identify the problems and to suggest policy developments for these parties as well as government. We compare perceptions of high-tech venture capital firms that invest in spin-outs with those that do not, and also consider VCs' views on spin-outs versus other high-tech firms. Our evidence identifies a mismatch between the demand and supply side of the market. In line with the pecking order theory, venture capitalists prefer to invest after the seed stage. However, in contrast to the pecking order theory, TTOs see venture capital as more important than internal funds early on. We develop policy implications for universities, technology transfer offices, academic entrepreneurs, venture capital firms and government and suggest areas for further research.

Keywords: Universities; Spin-offs; Venture capital; High-tech; Entrepreneurial finance

1. Introduction

The commercialisation of university activities has become a key part of the agenda for governments and universities. There has been a substantial rise in the creation of university spin-out companies (USOs)[4] (Lambert, 2003) as universities increasingly view equity ownership in a USO as an attractive alternative to licensing technologies in embryonic industries (Siegel et al., 2003a). In line with the Association of University Technology Managers (AUTM) in the US and other literature on spin-outs, we define a USO as a start-up company whose formation is dependent on the formal transfer of intellectual property rights from the university and in

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[4] We adopt the term spin-out but this is synonymous with the term spin-out used in US literature.

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which the university holds an equity stake. This definition is important with respect to the role that universities may play in venture development and the venture capital search process.[5] It means that we focus on those USOs that, in principle, may be expected to have high growth prospects but which may face difficulties in obtaining finance and other resources to realize these prospects.

The financial returns to this increased interest in knowledge transfer from universities have so far been low (Shane, 2004; Siegel et al., 2003a). Poor performance has led to an interest in understanding the potential difficulties associated with university commercialisation. The Lambert Review of University-Business Collaboration, commissioned by the UK government (Lambert, 2003), pointed to the problems associated with a dominant emphasis on the creation of spin-outs per se and argued that there was a need to focus on the requirements to enable the USOs that are created to generate significant wealth. This Review is in line with learning experience of IMEC[6] in Belgium (Moray and Clarysse, 2005) and Chalmers University in Sweden (Lundqvist and Hellsmark, 2003). The case of IMEC, for example, shows how the focus has changed from maximising the number of spin-outs towards optimising the starting configuration of these spin-outs in order to create the maximum future value.

The complexities of the problems involved in the development of USOs are well-recognised (e.g. Autio, 1997; Carayannis et al., 1998; Clarysse et al., 2005; Druilhe and Garnsey, 2004; Fontes, 2005; Mangematin et al., 2002). These studies also indicate that these complexities may be associated with the heterogeneity of USOs in terms of their resource endowments, their business models and their institutional contexts. Universities face a number of resource constraints in creating successful spin-outs but they cite access to venture capital as the most important (Table 1), with access to other forms of finance also figuring highly.

The existence of a gap between the demand for finance from entrepreneurs involved in new ventures and the willingness of suppliers to provide this finance has long been recognised in policy initiatives to help fill that gap in the US (Shane, 2004), UK (Rothwell, 1985; Bank of England, 1996, 2001, 2003) and Continental Europe (European Commission, 2000). Venture capital (VC) firms' provision of risk capital has been seen as a major solution to bridge the so-called equity gap for USOs. Venture capital firms can play a key role in enabling the development of new firms in new markets (Von Burg and Kenney, 2000) but venture capital firms in Europe especially have traditionally been criticised for being reluctant to invest in early stage high-tech investment (Murray and Lott, 1995; Lockett et al., 2002). The processes adopted by venture capital firms in screening potential investments have been widely researched (e.g. Shepherd, 1999; Zacharakis et al., 1999). However, the emergence of new ventures from universities, that have not traditionally been commercial environments, may introduce differences in the approaches adopted by venture capitalists. Commercialisation of activities through technology transfer is relatively new to many universities. Procedures for the realistic valuation of IP and its marketability are often poorly developed and designed (Leitch and Harrison, 2005). Understanding of the requirements of potential external funders may be quite low.

Although the culture and operating practices within universities may be changing in the context of technology transfer they are still very different from the private sector conditions familiar to providers of external finance. That problems arise in terms of communications between universities and external sources of finance is unsurprising and each needs to understand better the information and operational constraints to which the other is subject. For example, there has been growing recognition of the notion that ventures need to be in a pre-prepared state that enables venture capital firms to evaluate them more easily (e.g. Zacharakis et al., 1999; HM Treasury, 2001). As such, new ventures from universities, where the skills to prepare for venture capital investment are lacking, may face problems in attracting investment.

Shane and Stuart (2002) identify distinctions between USOs in terms of whether or not they are funded by venture capitalists. However, there has been a lack of analysis of demand and supply side issues concerning USO access to venture capital. Because of their importance for policies aiming to create wealth from the commercialisation of university research, this paper aims to fill this research gap by focusing on those ventures for whom venture capital may be appropriate but who may face problems in accessing it. For the purpose of this paper, we do not consider those USOs for whom venture capital is inappropriate, although these ventures may generate social and employment returns. We draw on research we have conducted in the UK, supplemented by our evidence from continental Europe, to address the following research and policy questions:

[5] In some jurisdictions, such as Sweden, the role of the university is much reduced as the intellectual capital belongs to the academic.

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Table 1
TTOs views on factors impeding the creation of university spin-out companies

Rank	Impediments	Mean value
1	Ability to obtain finance from venture capital firms	2.4
2	Availability of proof of concept funding for developing prototypes of the technology	2.5
3	The number of staff to manage the IP identification, assessment and exploitation process	2.5
4	Availability of proof of concept funding for conducting market research/analysis	2.6
5	Availability of proof of concept funding for conducting IPR due diligence	2.6
6	Incentives and rewards to attract commercial management to spin-outs	2.8
7	Availability of proof of concept funding for business plan development	2.8
8	Availability of resources to maintain patent applications	2.9
9	Ability to obtain finance from industrial partners	2.9
10	Ability to obtain finance from business angels	3.0

Source: authors' survey of TTOs. *Note*: institutions were asked to indicate the factors impeding or promoting the creation and development of spin-out companies during FY 2002 on a scale from 1 to 5 where 1, "strongly impeded"; 2, impeded; 3, "no effect"; 4, "promoted"; 5, "strongly promoted".

1. From the demand side, what are the problems faced by USOs in attracting venture capital finance?
2. From the supply side, what procedures do venture capital firms adopt when assessing USOs in terms of the stages at which they invest, the screening of proposals and the readiness of USOs for venture capital investment and to what extent are alternatives available?
3. What policies can be adopted by technology transfer offices, academic entrepreneurs, universities, venture capitalists and government to enable USOs to attract venture capital finance?

The paper is structured as follows. First we introduce a theoretical lens to analyse the equity gap and examine the policy instruments which have been developed to address this gap. Second, we outline the data and methodologies involved in our studies relating to market participants on the demand and supply sides. Third, we present new empirical evidence relating to the demand for venture capital. We then consider the supply side perspective of venture capital firms. We present new evidence from high-tech venture capital firms that provides insights as to why these firms may differentiate between investing in USOs from investing in other high-tech firms. We consider their investment preferences relating to the stages in which they invest, the screening of proposals and the issues involved in preparing spin-outs to the point where they are able to attract venture capital investment. Finally, we present conclusions and policy recommendations for technology transfer offices and academic entrepreneurs, universities, venture capitalists and government policymakers as well as implications for researchers.

2. Theoretical background

One of the dominant concepts to analyse the need for different forms of capital is the pecking order hypothesis (Roberts, 1991; Watson and Wilson, 2002). This hypothesis assumes that internal funding is preferred over external sources. Where these are insufficient, debt is preferred over equity in order to avoid ownership dilution. However, information asymmetries between the bank and the entrepreneurs may have negative consequences for the provision of bank finance and results in credit rationing (Stiglitz and Weiss, 1981). Evidence from the UK (Cressy, 1996) indicates that the constraint may not be as severe as initially accepted as some bank lending decisions reflect evaluations of human capital and collateral provision can be used to ameliorate the negative impact of information asymmetries.

In line with the pecking order hypothesis, we would expect that USOs, which are innovative start-ups facing severe financing constraints (Westhead and Storey, 1997), look for different forms of internal financing complemented with debt financing before they seek venture capital. Alternatively, we would expect that policy makers first make internal and debt financing easier to obtain before they introduce mechanisms to encourage venture capital as a way to stimulate spin-outs.

In the next paragraphs, we describe briefly the different forms of internal funding, debt financing and equity support that can be envisaged.

2.1. Internal funding

Internal financing is generated by the founding entrepreneurs themselves. US evidence on spin-outs suggests that internal funds provided by the entrepreneurs themselves accounted for upwards of 70% of initial capital (Roberts, 1991; Smilor et al., 1990). However, many university researchers in Europe may not possess enough financial sources to cover the start-up cost of such a venture. Therefore, many universities provide small amounts of funding to cover the costs of IP protection and very early stage developments. But even their finances would not normally accommodate large investments in commercialisation. In the financial year 2002, our survey detailed below, identified 33 universities in the UK that funded the development of USOs in this way.

2.2. Debt finance

The majority of smaller businesses tend to rely on bank debt rather than venture capital as their major source of external funds (Keasey and Watson, 1992; Scherr et al., 1993). Universities may be relatively well placed to put forward assets for the purposes of bonding debt finance. However, this would probably not be acceptable to generate funds for potentially high risk investments. An alternative is the use of the university's reputation or network as a quality guarantee, as well as visible indicators of technology quality such as patents and founding team track record.

2.3. Equity finance

Evidence indicates a limited role for bank finance in spin-outs in both UK and US studies (Roberts, 1991; Moore, 1994; EC, 2000a,b). Hence, equity providers such as business angels and venture capitalists might have to invest in the very early stages in the formation of a USO. Along these lines, Murray and Lott (1995) concluded that UK venture capital firms in the early 1990s were disinclined to invest in new technology based firms. On the other hand, Lockett et al. (2002) found that by the end of the 1990s this position was changing. While venture capital firms were becoming more flexible, there was continuing evidence that technology was seen as a more important risk factor than stage of investment. Asymmetric information problems in early stage technology ventures pose significant upfront search costs by potential financial providers.

Venture capital firms engage in detailed scrutiny of all aspects of potential investment proposals (Fiet, 1995; Muzyka et al., 1996; Shepherd, 1999). Due diligence is an essential part of the venture capital investment process designed to reduce various adverse selection problems arising from asymmetric information between the entrepreneur and the investor. The costs of due diligence discourage smaller investments that become prohibitively expensive in terms of evaluation and future monitoring commitments (Harvey and Lusch, 1995).

The implications of agency theory for the external financing of new and smaller businesses may be particularly significant where the activity involves new and untested technology involving products and processes where there are few previous data available in terms of marketability and profitability. These costs discourage participation in smaller investments and create what is often referred to as an 'equity gap' where formal venture capital is not available to projects below around £ 500,000. Lockett et al. (2002) found evidence that while technology-related deal flow appears to have increased over time, venture capitalists adopted different approaches to due diligence in examining technology and non-technology investments. VCs believed that many high-tech proposals continued to display greater shortcomings than non-technology projects regarding management quality, intellectual property protection and potential market size.

Although business angel investments may be made available in smaller amounts, such sources may not be appropriate in the case of university based IP (Mason and Harrison, 2004). Most angels appear to lack the requisite experience in high level science and engineering research and development to make a confident assessment of commercial potential.

2.3.1. Demand side conditions for equity financing

In seeking to access equity funding, university TTOs and the spin-outs they support may face a number of problems that, in theory, would be expected to be more significant than for other ventures. These may be attributed, first, to problems arising from the heterogeneity of their human capital in terms of the knowledge and understanding required to develop the case for investment in a way that is persuasive and meaningful to traditional venture capital suppliers.

It would also be expected that the IP in spin-outs would, by definition, refer to technologies and areas of specialism that are unfamiliar and possibly poorly understood in most traditional venture capital assessment processes. The very early stage nature of these developments would also make the case more difficult to construct even if the requisite expertise were available within the university. TTOs and entrepreneurs may need to understand the heterogeneity of the stage of technology development cycles and the scope and innovativeness of USOs' technology (Hindle and Yencken, 2004). Academic and university staff responsible for the preparation of such cases may find it difficult to predict the amount of funding required. They may also have unrealistic expectations in terms of the distribution of equity between participants and a poor understanding of investment readiness from a venture capitalist perspective.

Understanding and developing the business model of the USO, that is the way in which the resources are to be deployed to capture value, may be important in positioning ventures to attract venture capital. Different business models in different market sectors may give rise to heterogeneity in the types of spin-outs (Druilhe and Garnsey, 2004). This suggests a need to undertake activities to establish the appropriate market positioning of the spin-out.

The institutional context may also influence demand side factors as heterogeneity in universities' objectives and support processes may affect the kind of spin-outs emanating from universities that may be attractive to venture capitalists as well as the resources and incentives made available to develop spin-outs (Radosevich, 1995; Debackere, 2000; Lundqvist and Hellsmark, 2003; Clarysse et al., 2005).

2.3.2. Supply side constraints on equity financing

The information asymmetries that raise problems for those applying for finance would also be expected to have a similar impact on supply side conditions. Since USOs emanate from a traditionally non-commercial environment these problems facing venture capitalists may be especially stark. The venture capitalist's emphasis upon the management team may be compromised by human capital shortfalls as indicated above. The early stage and unpredictable nature of many of the technologies may not be easily accommodated by the existing human capital stock within most venture capital institutions. The academic scientist may not be the most appropriate CEO and there may be problems in finding suitable alternatives (Franklin et al., 2001). It would also be expected that risk assessment measures are more difficult to apply due to high levels of uncertainly surrounding the efficacy of the technology and the marketability of its functionality. A particular issue concerns the nature of IP protection. Following the UK Patents Act 1997, the IP may not be owned by the academic inventor but by the university. The nature of individual universities' objectives, strategies and support for commercialisation may affect the ability of venture capitalists to negotiate an appropriate deal that would enable them to achieve their target rates of return (Kassicieh et al., 1996; Seashore Louis et al., 1989; Samsom and Gurdon, 1993). VC firms themselves may be heterogeneous in terms of their funding source, their objectives, their human capital and financial resources and their geographical scope.

In general, therefore, it would be expected that the issues surrounding both demand and supply conditions resulting from the impact of asymmetric information would create higher levels of market failure in the area of university IP and technology transfer through spin-outs.

3. Policy and context

3.1. Government policies

The problems relating to the provision of equity finance for new ventures have been long-recognised in Europe (Murray and Lott, 1995; Delapierre et al., 1998; CEC, 2003). Since the 1990s, various government initiatives have been implemented in several European countries to stimulate funding for new high-tech firms (Cieply, 2001; Pollock and Scheer, 2002; Heydebreck et al., 2000). The different actions can be divided into six types of measures: (1) 100% publicly owned funds

Table 2
Types of programme

Type of programme	Description	Example
Public fund	100% publicly owned funds focused on pre-seed and/or seed stages	Twinning Growth Fund and Biopartner (The Netherlands)
Public/private equity fund	Fund in which government and private sector co-invest with same focus as previous	University Challenge Funds (UK); Technologiebeteilungesellschaft (Germany)
SBIC-type of refinance schemes	Schemes which leverage the deals made by adding public money to the private investment	Arkimedes (Belgium); SBIC (USA); Kreditanstalt für Wiederaufbau (Germany)
Guarantee schemes	Insurance schemes which guarantee (part) of the VC money in case of bankruptcy	Various, in each country
Fiscal incentives	Tax reduction scheme on value added or income tax deduction	Aunt Agaath Scheme (Nl); Banque de Développement des PMEs (France); Trust Capital Funds (UK)
Incubation scheme	Scheme which pays the salaries of coaches, which offers facilities and/or which offers network opportunities	National Incubator Programme (Sweden); DIILI Programme (Finland); Innovationsmiljoer (Denmark); Exist (Germany); Les Incubateurs Publiques (France)

focused on pre-seed and seed stages; (2) public–private partnerships with the same focus as the previous; (3) small business investment companies (SBIC)-type of refinance schemes; (4) guarantee schemes (risk sharing); (5) fiscal incentives in the form of tax reduction on value added or income tax deduction; (6) incubation schemes (Table 2). Some of these schemes are common across Europe while others relate to particular countries.

Some governments have created public funds with the specific aim of financing (high-tech) spin-outs at a very early stage. These funds only have public parties as institutional investors and are therefore publicly owned, such as the Finnish fund "Sithra". This fund has a budget of € 235 million and is involved in *pre-seed*, *seed* and *early stage venture capital*. In the pre-seed and seed stage, additional coaching is given to improve business plans and finance market research activities. Other countries that have created 100% public funds to finance the pre-seed and seed stage include The Netherlands (Twinning Growth fund, Biopartner), Denmark (Danish Growth Fund) and France (Fond de Co-investissement des Jeunes).

A second category of policy initiatives involves the establishment of public–private partnerships set up to finance (high-tech) start-ups. Public investment in these funds varies from 10% to over 90%. Almost every European country has its own public–private partnership fund. A specific category of such partnerships are the university funds. Pressure from government on universities to commercialise IP (HM Treasury and DTI, 2004) has been accompanied by the establishment of various programmes to facilitate the process. Usually, the establishment of seed and pre-seed funds associated with the university are part of this process. University funds almost always invest exclusively in start-ups that are based on technology developed at the parent institution with which they are associated. The degree of public involvement varies from country to country. Programmes such as University Challenge Fund (UCF) in the UK comprise up to 77% public capital, while the University Seed Funds in Belgium (Flanders) only have 20% public capital, on average.

A third, much less popular, set of initiatives is inspired by the US SBIC[7] scheme. In these schemes, the VC fund that invests in a (high-tech) spin-out can co-finance its investment by a loan which is granted by the government. As the loan is to be reimbursed at a fixed, usually below market rate, the scheme increases the potential profit that can be made on such an investment. It does not, however, decrease the risk for the venture capital fund. In Europe, only Germany (Kreditanstalt für Wiederaufbau) and Belgium (Archimedes) have such programmes in place.

The fourth category of policy measures aims to reduce the risk to the venture capital firms through guarantee schemes. If a start-up fails, the government can reimburse the venture capital firm which qualifies for the scheme for (part of) its lost investment. There is no consensus about the effectiveness of these schemes since some tend to argue that they give an incentive to VC funds to let their portfolio companies go bankrupt in order to recover at least part of their money invested instead of trying to save these companies. Despite this lack of consensus, such a guarantee scheme exists in almost every European country.

A fifth form of policy measure comprises "fiscal incentives". They provide tax reductions on capital gains created by investments or they allow investors in venture capital funds that qualify for the scheme to offset (parts of) their investment against taxes. In the UK, VC Trusts and the Enterprise Investment Scheme provide private investors with tax relief on income and capital with venture capital firms involved in managing the trusts. In The Netherlands, private investors that invest in small business created by relatives receive a significant deduction in income taxes through the Tante Agaath Scheme. In France (Banque de Développment des PMEs) and in Belgium (Privak system), the surplus value on shares invested in venture capital companies that qualify specific criteria is subject to a much lower tax scheme.

Finally, a sixth form of support refers to business incubation schemes. In the late 1990s, the governments in France and Sweden launched their National Incubation Programmes (Jacob et al., 2003). Also the German EXIST programme and the Danish Innovationsmiljoer are incubation like initiatives. In these initiatives, governments usually do not provide direct pre-seed or seed capital, but provide infrastructure and management support for high-tech start-ups. Often, the incubator is linked to a university or public research institute.

The categories described above are not always mutually exclusive but they provide a useful framework to discuss the various initiatives which have been taken by European governments to stimulate the financing of high-tech start-ups.

3.2. Venture capital provision

From accounting for only 5.5% of investments made by British Venture Capital Association (BVCA) members in 1996, early stage technology investments grew rapidly to account for 30.4% in 2000. While this level

[7] Small Business Investment Companies.

of activity fell following the dot.com boom, early stage technology investments continued to account for around a quarter of all BVCA investments during the period 2000–2003 (BVCA, 2004). The value of early stage VC investments in technology followed a similar pattern, rising from 3.3% of total BVCA investments in 1996 to reach 7.7% in 2000 and falling to 5.5% in 2003. A similar pattern is evident at the European level. After accounting for 18.2% of investments made by EVCA members in 1996, early stage investments accounted for 46.8% in 2000 and 34.7% in 2003. In 2003, they amounted to 7.4% of total investment value as against 19% in 2000 (EVCA, 2004). There is, however, some variation between venture capital markets in the share of investment volume accounted for by early stage investments. At the lower end of the spectrum, only 16.6% of venture capital investments in The Netherlands in 2003 involved early stage deals while at the upper end, some 42.7% of venture capital investments in Germany involved early stage ventures (EVCA, 2004).

The trend in returns to technology investments may also reflect problems in this segment of the venture capital market. Mean internal rates of return on technology investments rose from 9.3% in 1997 to 12.8% in 2000, before falling to 7.4% in 2003. Over the same period, mean internal rates of return for non-technology investments rose from 14% to 17.3% before declining to 14.5% (PwC/BVCA, 2004). Across Europe, the median IRR for all venture capital investments (excluding buy-outs) for funds formed over the period 1980–2003 at the end of 2003 was 0.0% and the top quartile IRR was 7.4%, while the corresponding figures for all later stage private equity funds were 0.0% and 10.9%, respectively (EVCA/Thompson Venture Economics, 2004).

The context of our studies on the provision on the impact of venture capital on USOs is, therefore, one in which governments are actively developing public programmes to stimulate spin-out activity and venture capitalist interest in spin-outs. However, as we have shown, there remain some differences between countries in the extent and nature of this development. At the same time, although venture capitalists have become more willing to invest in high-technology related investments, the returns associated with such investments have traditionally lagged non-high-technology investments.

4. Data and methodology

To examine issues relating to spin-out access to VC, we surveyed university technology transfer officers, venture capitalists and academic entrepreneurs in the UK which we supplement with evidence from continental Europe. We present this evidence, from a number of different perspectives, in order to *triangulate* the interviewees' views on issues relating to spin-outs and venture capital (Deshpande, 1983). We expand on the data collection processes and methods below.

4.1. Demand side

Building on existing literature and qualitative interviews with a small number of TTOs and representatives of spin-outs, we developed a questionnaire instrument in order to survey university technology transfer activities. In March 2003, this questionnaire was sent to the top universities in the UK as ranked by research income, accounting for 99.8% of this revenue. We identified the most suitable respondent through UNICO and AURIL membership as, typically, the head of the technology transfer office or their designate. We received full information from 124 of the 125 universities who were members of AURIL and UNICO. The remaining 46 universities and higher education institutions were not contacted as these accounted for 0.2% of research grants and contract expenditure and hence were not involved in commercialisation activity.

The quantitative survey of TTOs was followed in 2003 and 2004 by a series of detailed interviews with a selection of TTOs in order to enable insights to be gained regarding the processes used to get projects investor ready. We approached TTOs in eleven institutions within the context of maintaining coverage that reflected a range of age, experience, geographical spread and size. The universities in our sample are drawn from a wide range of geographic regions in the UK. The universities also display a substantial age range. The sample includes both some of the UK's longest-established universities as well as those more technologically focused universities created during the 1960s.

To supplement the UK data, the same questionnaire was used to assess spin-out activity in a sample of 50 universities located in different regions throughout Europe. We selected eight universities in six European regions[8] that belonged to the top research intensive regions in Europe according to the European Commission (Third Report on S&T Indicators, 2004). The universities within the regions were selected using a stratified approach with the aim of maximising their representativeness in terms of research policy. Universities with a specific research

[8] Midlands (UK), Stockholm (Sweden), Île de France (France), Baden-Württemberg (Germany), North and Central Italy (Italy), and Helsinki (Finland).

focus were matched to those with a general one. In addition, medical related institutes were matched with those where ICT was the main target of research. In each of the four categories of this two by two matrix, two institutes were selected which had a track record of at least eight spin-outs. For each university, eight spin-outs created after 1991 and before 2001 were randomly selected for further interviewing.

4.2. Supply side: venture capitalists

Given our research questions, we focus on venture capitalists that indicate an interest in investing in high-tech, early stage deals. This very specific population of VCs includes firms with varying degrees of public capital in their shareholder structure. The majority of VC firms that are active in these early stages leverage their capital or downsize their risk usually by some form of public involvement as noted earlier. This can be through straight co-investment schemes at national or European level. Alternatively, fiscal SBIC-type of incentive schemes can be used to collect money by the VC funds or to leverage profitability on a deal specific basis. The complexity of the public–private relation makes it impossible a priori to focus on a sub-population of public or private funds.

In late 2003, we undertook a study of venture capital and its provision to spin-outs. Questionnaires were sent to the 56 venture capital funds in the UK that identified themselves as being active investors in technology based small firms as indicated by the BVCA definition (BVCA, 2004). We aimed to examine the attitudes and perceptions of venture capital investors and to analyse the factors affecting the supply of finance from venture capital funds for spin-outs. We received 27 fully completed questionnaires plus nine nil responses/partially completed returns; i.e. we obtained usable responses from fifty per cent of the active VCs in the high-tech market. This response rate is in line with other surveys of VCs in the UK and very encouraging for a mail survey (Lockett et al., 2002).

Again to supplement the UK data, we conducted face to face interviews with 65 VC firms located in the six European countries noted above[9] which mentioned in their local venture capital association directories that they invested in high-tech start-ups. To maximise the intra-sample variety of these VC firms, the sample frame of VC firms was divided into four groups: funds with more than 50% public capital, captives which belong 100% to a financial institute, multinational VC firms and local VC firms. In each category a random selection of four firms was made. From the 96 identified VC firms, 65 (60%) agreed to participate in the study.

[9] Because of the concentration of the financial capital in Germany in the Bayern region instead of Baden-Württemberg, we replaced the latter by the former in the VC-study.

5. Empirical evidence

In this section we present evidence initially in terms of consideration of the extent of VC and other equity funding of spin-outs. We then consider the importance of external finance from the demand side perspective of the spin-out *and* the universities involved in their creation. This is followed by the supply side perspective of the VC firm in terms of their investment preferences relating to the stages in which they invest, the screening of proposals and in terms of the issues involved in preparing spin-outs to the point where they are able to attract to VC investment.

5.1. External investments in spin-outs

In terms of *new* companies attracting equity finance, notwithstanding the fact that University Challenge Funds (UCF) may not be accessible to all institutions, this is the most popular source of finance (Table 3, Panel A). In total 50 investments were made in spin-outs across the 113 universities providing data. The second most popular source of finance was VC with 20, followed by industrial partner (14) and business angels (12) investments made respectively. In terms of *existing* spin-outs, that is companies established prior to financial year 2002, VC finance was the most popular source of funds with 42 investments made, followed by UCF (32), business angel (21) and industrial partner (5) (Table 3, Panel B).

Using the same questionnaire in Continental European universities, we find consistent evidence with our UK results. In Continental European universities UCF equivalent finance is mainly a substitute for internal funds of universities or public research institutes. In those countries where no UCF kind of programme exists, such as Germany, the *pre-seed* capital is invested by the university itself.

These findings are not in line with the pecking order theory, which seems to suggest that the founders' own savings would be the most important source, followed by debt financing. In fact, TTOs view these two sources as not important at all, albeit the top universities find the investment of personal funds to be significantly less important than those universities without a track record in spinning out companies. In those universities, it seems that challenge fund instruments are substituting for the lack of internal funding.

Table 3
TTOs views on importance of sources of finance for spin-out companies

Source of finance	Combined mean	Top performers mean	Non-top performers mean
Venture capital	2.8** (2.7)	2.2 (1.8)	3.1 (3.2)
Government grant (e.g. SMART awards)	2.2 (3.0)	2.2 (3.2)	2.1 (2.9)
University's own funds	3.2 (3.3)	3.2 (3.3)	3.3 (3.3)
Joint ventures between the university and an outside firm	2.8 (3.3)	3.2 (3.1)	2.6 (3.4)
University Challenge Fund money	2.8** (3.5)	1.8 (2.1)	3.3 (4.2)
Business angels	3.0 (3.5)	2.9 (3.1)	3.0 (3.7)
Licensing deals	3.0 (3.8)	3.2 (3.7)	2.9 (3.9)
Founder's own savings or capital	3.9*** (4.1)	4.6 (4.2)	3.5 (4.0)
Bank debt	4.1 (4.4)	4.5 (4.4)	3.9 (4.3)

Source: authors' survey of TTOs. *Note*: respondents ranked each source of finance as: 1, "extremely important"; 2, "very important"; 3, "important"; 4, "somewhat important"; 5, "not important". A Mann–Whitney test was performed to analyse the differences between the top performers and the non-top performers. Top performers are defined as those universities having created more than five spin-outs in the period.
** 1% significance level.
*** 0.1% significance level.

In addition to the importance of different forms of finance, Table 3 also indicates that spin-outs that attract external equity investments tend to be skewed across a small number of universities. In particular, for new spin-outs only 16 universities had companies that attracted VC investment, 11 business angel investments and 12 industrial partner investment. Investment in existing spin-outs is also highly skewed, with only 17 universities attracting UCF finance investments, 24 attracting VC investments, 12 accessing business angel investments and 13 finding industrial partner investments for these ventures.

In summary, although external equity finance is deemed very important to the development of spin-outs, the distribution of spin-outs, and in particular, equity-backed spin-outs is highly skewed across UK universities. In terms of formal VC – that is, excluding UCF, business angel and industrial partner finance – across the 113 universities providing data, only 62 investments were made in both new and existing spin-outs in the financial year 2002. The following sections seek to shed light on the reasons for this level of activity.

5.2. Demand side

Case study analysis of USOs indicates that they face a number of critical junctures in their development (Vohora et al., 2004) that may distinguish them from other new ventures. Access to VCs may be especially important at the opportunity recognition and credibility junctures. Developing this analysis with new evidence from TTOs, we find that those universities most active in spinning-out ventures are significantly more likely to recognise the importance of access to VC (see Table 4). Indeed, for these universities, after UCF finance, venture capital is seen as the second most important source of finance ahead of business angels and industrial partners.

Our evidence from university TTOs, noted earlier (see Table 1) also indicates major problems in developing prototypes, conducting initial market research and in developing business plans because of shortages of proof of concept funding. A lack of adequate human resources is also indicated with respect to the availability of staff to identify, assess and exploit intellectual property. Incentives and rewards for such staff also appear to pose problems. This evidence suggests some recognition that the wider constraints and complexities in developing new ventures create problems in getting spin-outs ready for VC.

5.3. Supply side: venture capitalists' views on spin-outs

5.3.1. Proposals and investments

Our VC respondents reported they received investment proposals from six universities on average and spent time actively engaged in developing business plans with an average of 3.8 universities. Of the 27 responding VC firms, only eight made USO investments with at least four different universities. Only three VC investors funded more than one USO from the same university.

In our face to face interviews with the 65 VC firms, located in six different European countries noted above, 41 explicitly mentioned their willingness to invest in USOs, while only 15 do not consider such an investment and eight were undecided.

Investors preferred to invest in both USOs and high-tech companies at their start-up stage where there is a lower level of risk than at the seed stage at which proof-of-concept work occurs. At all stages of investment,

Table 4
Sources of external finance for new and existing spin-outs FY2002

No. of investments made by university	No. of universities making investments with each source of finance			
	UCF	VC	BA	Corporate partner
Panel A: equity investments in new spin-outs FY2002				
0	90	97	102	101
1	7	12	8	10
2	9	4	2	2
3	4	–	–	–
4	2	–	–	–
5	1	–	1	–
Total number of investments made	50	20	12	14
Panel B: equity investments in existing spin-outs FY2002				
0	96	88	101	100
1	9	17	8	1
2	5	3	2	2
3	1	2	1	–
4	1	1	–	–
5	–	–	–	–
6	1	2	1	–
Total number of investments made	32	42	21	5

Source: author's survey of TTOs. *Note*: UCF, University Challenge Fund; VC, venture capitalist; BA, business angel; IP, industrial partner.

more VC firms were willing to invest in high-tech companies than in USOs. This is in line with the pecking order theory.

Proposals for start-up stage high-tech investments accounted for almost three quarters (74.3%) of all investment proposals received (Table 5, Panel A). Across all stages, 90.2% of investment proposals received were for high-tech ventures and only 9.2% were for USOs.

The 27 respondents made a total of 60 investments in high-tech companies and 43 investments in USOs in calendar year 2002 (Table 5, Panel B). Start-up stage high-tech investments accounted for almost 40% of all investments by number. This category of investment also received the greatest number of investment proposals.

However, further analysis provides more fine-grained insights beyond this accepted view. A higher proportion of seed stage investment proposals may be attractive to USO investors than start-up stage investment proposals as the ratio of investments to proposals is highest in respect of seed stage spin-out ventures (Table 5, Panel C). For USOs, the ratios of investments made to proposal ratios are higher at both seed and start-up stages. Interestingly, these figures indicate that at both these stages, USO proposals stand a reasonable chance of leading to investment alongside similar investment proposals. These ratios also show that investment proposals for USOs stand a better chance of receiving funding at the start-up stage than at the seed stage from VC firms.

Table 5
Venture capitalists investment proposals and investments

	Seed stage		Start-up stage		Late stage	
	High-tech	USO	High-tech	USO	High-tech	USO
Panel A: proposals						
Total number of proposals	815	369	4599	135	201	69
Percentage of total number of proposals (%)	13.1	6.0	74.3	2.2	3.2	1.1
Panel B: investments						
Total number of investments	15	24	41	18	4	1
Percentage of total number of investments (%)	14.6	23.3	39.8	17.5	3.9	0.9
Panel C: investments to proposals						
Ratio of investments to investment proposals	0.018	0.065	0.001	0.13	0.02	0.014

Source: authors' survey of venture capital firms.

Table 6
Venture capital funds invested by stage of investment

	Early seed stage		Early start-up stage		Late stage	
	High-tech	USO	High-tech	USO	High-tech	USO
Total amount invested (£)	12970000	6786000	24750000	7867000	5500000	100000
Percentage of total amount invested (%)	22.4	11.7	42.7	13.5	9.5	0.2
Average amount invested (£)	865000	283000	604000	437000	1375000	100000

The average minimum preferred size of investment was £ 650,000 and the average maximum preferred size of investment was £ 2.15 million. This range suggests that seed/proof of concept stage deals may be too small for many venture capital firms to consider because of the transaction costs involved. For VC firms, the transaction costs involved relate to the costs of screening potential investees prior to invest, together with post-investment monitoring costs. In absolute terms nearly three quarters (74.6%) of the total amount invested went to high-tech firms and the majority of this was invested at the start-up stage (Table 6). When considering the average amount invested according to type of investment, VC firms have invested significantly smaller amounts in USOs investments at each stage.

5.3.2. Screening of proposals

Differences in screening processes leading to proposal rejection between those VCs who are interested in USOs, and those who are not, highlights the issues that TTOs and USOs may need to address if they are to persuade venture capitalists to invest in their ventures. The significant differences between VC firms who are willing to invest in USOs and those who are not in terms of the criteria they use to reject an investment proposal are ranked in order of importance in Table 7.

Non-USO investors place greater emphasis on the size of the potential market when considering investment proposals. The difficulty many universities have in commercialising disruptive technologies emerging from basic research is that of identifying markets in which to apply the technology and estimating the level of demand. This result is confirmed in the findings from the 65 VC firms spread over different European regions. Non-USO investors are significantly more concerned about market size, which can provide them with a significant exit opportunity. USO investors on the contrary seem to be more concerned about the economic viability of the venture and find the estimated time to break-even a major point of importance.

Joint ownership of IPR with a university is significantly more important to non-USO investors. Investors feel uncomfortable about investing in USOs when IP is licensed as opposed to being assigned in return for

Table 7
Venture capitalists views on factors leading to rejection of investment proposals: main significant differences between spin-out investors and non-spin-out investors

Factors of significant differences between spin-out investors and non-spin-out investors	Combined score	Non-spin-out investors	Spin-out investors
Size of potential market for applications of the technology	4.2***	4.6	3.9
Stage of development of the product/service	4.1**	4.7	3.6
Availability of a prototype/test data to demonstrate proof of concept	3.5**	4.6	2.8
Difficulty in identifying key decision makers	3.4#	4.1	3.0
Lack of formalised university technology transfer procedures	3.3*	3.9	2.8
Requirement for service development to support customers who will use the product/service	3.0*	3.7	2.4
Concerns over co-investing with public sector funds	2.9*	3.7	2.4
Concerns over co-investing with universities	2.8*	3.4	2.3
Joint ownership of the IPR with universities	2.6**	4.0	1.9

Source: authors' survey of venture capital firms. *Note*: respondents scored each factor as: 1, "unimportant"; 2, "not very important"; 3, "quite important"; 4, "important"; 5, "very important". A Mann–Whitney test was performed to analyse the differences between spin-out and non-spin-out investors.

* 5% significance level.
** 1% significance level.
*** 0.1 significance level.
\# 10% significance level.

an equity share in the company. Some investors believe that separation from the university is necessary for the spin-out to develop without interference. In the event of failure, the IP may be the only valuable asset that can be recovered and sold off. Universities often prefer to retain control over IP to USOs as they wish to retain some control over IPRs as well as expecting an equity stake in lieu of royalty payments. In this way, non-USO investors may perceive universities as wishing to retain ownership and control without sharing the risk involved.

Non-USO investors place more emphasis upon the need for a prototype in order to assess the viability of the technology concerned. In contrast, USO investors may invest at an earlier stage and work with the USO to achieve proof of concept. Without adequate resources to achieve proof of concept it is unlikely that proposals for USOs will meet the investment criteria of non-USO investors. This finding is also in line with our European sample of 65 VC firms. USO investors in this sample departed from the idea that a good contact with the entrepreneur and a protected technology are good starting points for an investment. Both the development of the technology into a product and the development of a viable management team is in their view something that could be accomplished afterwards. In contrast, non-USO investors put much more emphasis on a working prototype and the availability of a professional management team in place before the investment is made.

Difficulties in identifying key decision makers in universities and a lack of formalised university technology transfer procedures are also a significant source of discouragement for non-USO investors. Negotiation procedures with universities and the possibility of their future participation as co-investors are also problematical issues.

Table 7 shows only significant differences between the two groups of investors. We also found that while the level of patent protection was important in the decision to reject a proposal for both groups (mean score 4.0), the difference between them was not significant. Similarly, there were no significant differences between the two groups in terms of the skills of the entrepreneurial team (mean score 3.8), whether there was a clear route to market for the technology (mean score 4.5), policy not to invest in certain sectors (mean 3.8), familiarity with certain technological markets (mean 3.4) and difficulties in raising finance for certain sectors (mean 2.9).

There were, in general, common perceptions between the private sector VC investors in the sample and those with public sector involvement. The only significant differences related to those VC firms with public sector involvement being more likely to reject USO proposals because of problems with the business development skills of the management team (public mean score 5.0, private mean score 3.7) and because of the overall quality of the proposal (mean 5.0 versus 4.5). Conversely, VC firms with public sector involvement were significantly less likely to reject USO proposals because of the stage of development of the product (mean 4.0 versus 4.5) and because of concerns about co-investing with business angels (mean 2.0 versus 2.5). We also compared larger and smaller VC firms (defined as those firms above or below the median funds managed) and found no significant differences on these variables.

Table 8
Venture capitalists attitudes and perception of risks associated with investment in USOs

Rank	Compared to high-tech companies, USOs are more likely to	Mean
1	Require building a management team	4.4
2	Require a longer investment time horizon	4.3
3	Require close monitoring	4.2
4	Require several rounds of funding	4.2
5	Have higher variability of return	3.6
6	Fail	3.6
7	Involve protracted pre-deal negotiations	3.5
8	Be small niche market companies	3.3
9	Pose valuation difficulties	3.2
10	Have financial structuring problems	3.1

Source: authors' survey of venture capital firms. *Note*: respondents ranked each factor as: 1, "strongly disagree"; 2, "disagree"; 3, "neither agree or disagree"; 4, "agree"; 5, "strongly agree".

In the European sample, we found that those venture capitalists with a higher level of public funding invested in earlier stage deals and looked more at the technology, whereas those with lower levels of funding focused more on traditional venture capital criteria such as market size and founding team structure.

A second dimension of screening that TTOs and USOs need to address relates to VC investors' perceptions of risk and reward when considering investments in spin-outs as compared to other high-tech companies. The most important issue where potential investors considered there were greater risks in USOs involved having to build management teams (see Table 8). This quantitative evidence relates to the quality of management and is consistent with our qualitative evidence that academics may not have the credibility to attract customers or to attract management with commercial expertise.

The time taken to realize substantial returns, and the prospect of multiple funding rounds before an exit can be achieved, make investment in spin-outs less attractive. The low level of exit was clear from evidence provided by the VC firms. In 2002, the VC firms surveyed had only exited 14 investments fully and four partially. The most common form of exit was liquidation (10). There were also four full exits and a trade sale to another corporation. Problems of early and efficient exit are a reflection of the early stage of development of most university technologies and the lengthy product development cycles in sectors such as life sciences.

The perceptions of risk and reward from spin-outs were generally common between the private sector VC investors in the sample and those with public sector involvement. The only significant difference related to those VC investors with public sector involvement perceiving greater problems in valuing spin-outs (mean score 4.3) than private sector venture capitalists (mean score 3.1). We also compared larger and smaller VC firms (defined as those firms above or below the median funds managed) and found no significant differences on these variables.

5.3.3. Investor readiness

Spin-outs need to be matched up to acceptance criteria expected by investors. As shown in Table 9, those universities that were most active in spinning-out ventures were significantly more likely to identify the importance of various processes in meeting these criteria when promoting USOs.

Our interviews with TTOs also indicated a number of procedures that were undertaken to render ventures investor ready. These included the provision of help with business planning, finding managers and assisting with fund raising. More active TTOs had informal networks to find managers and budgets to pay for consultancy, academic secondments and the purchase of specialist equipment.

Most TTOs were involved with IP assessment and due diligence. Some universities made use of patent attorneys while others had the expertise in-house. There were some indications that sufficient due diligence was not being undertaken and that too much reliance was being place on the motivation of the academic.

5.3.4. Venture capitalists' experience

It is also important to investigate the skills of VC executives to screen and manage investments.

Table 10 reveals that the most common form of experience for VC executives was in the financial services sector (46%), followed by accountancy (31%), non-executive or executive director of a technology firm (30%), graduate qualification in a tech subject (29%), managerial experience in a tech firm (28%), Ph.D. in a technology related subject (14%) and law (12%). Of the 25 firms active in the high-tech VC market only 14% of investment executives (averaged across the firms) had a Ph.D. in a technology related area. This low proportion may raise issues relating to how well VC executives are able to understand the science on which the spin-out companies are formed. This problem may be exacerbated since only 29% of investment executives (averaged across the firms) had a graduate qualification in a technology related subject. Assuming that all science Ph.D.s have a graduate degree, this means that over 70% of high-tech investors do not have a formal science education to a level higher than that taught at high school. This evidence also suggests that VC firms even in the high-tech sector are heterogeneous in terms of their expertise.

Again, this problem is present across Continental Europe. The experiences of the investment managers in the 65 European VC firms interviewed were very

Table 9
TTOs views on factors promoting the creation of university spin-out companies

Impediments	Combined mean	Top performers mean	Non-top performers mean
Incentives and rewards for institution staff to spend time on IP exploitation activities	3.2	3.5	3.0
Internal processes for conducting business development	3.2**	3.7	2.9
Internal processes for spinning out new companies	3.3	3.7	3.1
Internal processes for conducting IPR due diligence	3.3**	3.7	3.1
The level of marketing, technical, negotiating skills of staff involved in IP exploitation activities	3.4***	3.9	3.2
Incentives and rewards for academics to exploit the IP generated from their research	3.5**	4.00	3.3

Source: authors' survey of TTOs. *Note*: institutions were asked to indicate the factors impeding or promoting the creation and development of spin-out companies during FY 2002 on a scale from 1 to 5 where 1, "strongly impeded"; 2, impeded; 3, "no effect"; 4, "promoted"; 5, "strongly promoted". A Mann–Whitney test was performed to analyse the differences between the top performers and the non-top performers.

** 1% significance level.

*** 0.1% significance level.

Table 10
Venture capital executives' experience

VC experience/background	Mean percentage of a firms executives	S.D.
Panel A: high-tech VC executives' backgrounds		
Graduate qualification in tech subject	29	37.21
Ph.D. in tech subject	14	25.31
Managerial experience in a tech firm	28	39.07
Managerial experience of financial sector	47	36.39
NED or director of a tech firm experience	30	40.79
Chartered accountancy	31	34.48
Law	12	23.07

Percentage of VC firms with investment executives with Ph.D. in technological subject	Frequency
Panel B: Ph.D. in technological subject by VC firms	
0	18
1–10	1
11–20	–
21–30	–
31–40	1
41–50	1
51–60	2
61–70	2
71–100	–
Total number of high-tech VC firms	25

Source: authors' survey of high-tech VC firms.

similar to the ones reported by the British VC firms. Moreover, only in the VC firms with an explicit biotech focus were Ph.D.s found. VC funds with an explicit ICT focus, employed a disproportionate percentage of managers with a financial background.

Panel B of Table 10 also indicates that technology related experience is highly skewed towards a small number of venture capital firms even amongst that subset of the whole VC industry investing in high-technology ventures. Table 10, Panel B shows that 18 firms (of the 25) have no investment executives with a Ph.D. in a technology related subject. Therefore, not only is there arguably a low level of formal technology related education across investment executives but this experience is concentrated in only a small number of firms. This information raises interesting issues for the VC community.

5.3.5. *Venture capitalists and alternative investors*

Our discussion raises the issue of the extent to which alternative investors provide a substitute for VC investors both in the provision of finance and other resources and capabilities.

A central problem for universities in developing links with business angels who have backgrounds in the technology area (Mason and Harrison, 2004), lies in finding partners with the appropriate industry and technology background. It is debatable whether the highly successful entrepreneurs necessary as partners to stimulate high growth ventures are to be found among the members of local business angel networks. University TTOs might envisage building up a European level network of business angels in those technologies in which the university is strong. Alternatively, the European Association of Business Angel Networks could play a more active role in this for those universities that do not have the critical mass to build up such a network.

A second alternative concerns corporate investors. Our interview evidence relating to USOs involving disruptive innovations where the university had acquired patents suggests that in comparison to the venture backed USOs, USOs involving joint ventures with corporations derived resource benefits from the prior knowledge of their industries and the superior market intelligence held by the corporate partner. As such they may be better positioned to develop opportunities from scientific discoveries, which better serve unmet customer needs. In these cases, the corporate partner and the university may be better able to cooperate in assembling a well balanced team with the necessary background to successfully commercially exploit the technology. The corporate partner may be able to use its reputation to attract experts in product development and manufacturing to work for the new venture. Corporate partners can provide greater access in-house to resources such as marketing, technology, raw materials, equipment, facilities, finance, managerial expertise and organisational routines and control systems. The credibility acquired from both the university and the corporation may enable these ventures to acquire further resources more readily. The transfer of knowledge from the parent corporation may also produce more rapid professionalisation and capability building that enables them to enter and compete in new markets. The stock of social capital developed by corporate partners from their long-term involvement in a sector can help expedite the development process.

Venture capitalists may be able to provide similar support if they acquire the relevant social capital by recruiting executives with experience in a particular sector or by building a network of consultants and non-executive directors. From our interviews with VC firms, there was evident heterogeneity in their ability to develop USOs. For example, one VC was UK focused with limited expertise in growing high-tech firms and with limited networks of business contacts. This created problems for the USO in which it invested, with there being irregular contact between the VC and the USO, little advice on strategy and an inability to identify customers, managerial expertise or further finance. In contrast, a larger international early stage VC we interviewed had extensive expertise in growing high-tech firms in different industries and extensive social capital. It was able to provide closer involvement to develop the USO.

6. Conclusions and recommendations

Our analysis has enabled us to identify both the problems involved in attracting VC for USOs and their potential solutions from different perspectives. Our results indicate that in contrast to the pecking order hypothesis, TTOs and USOs see venture capital as more important early in the process than internal funds. Our findings also indicate that the majority of VC investors surveyed prefer to invest in USOs *after* the seed stage, particularly once proof of concept has been achieved; this is in line with the pecking order hypothesis. We also found that at both seed and start-up stages, USO proposals had a greater chance of receiving funding than non-USO high-tech investments. This relatively high acceptance rate for USO proposals compared to high-tech proposals may be indicative of screening and preparation being undertaken by technology transfer offices.

Surprisingly, although universities point to the difficulties in attracting VC and business angel money, they seldom develop mechanisms to increase the possibility of internal funding and debt financing as an intermediate step towards VC investment. This is no doubt due to the financial constraints which universities face. It is only among the top universities in the UK that the almost 100% public challenge funds can be seen as a real substitute. Furthermore, the lack of debt financing as an intermediate step is notable. Moreover, most countries have developed guarantee schemes so that the need for collateral is backed by public money. We feel that it is important for TTOs to invest much more effort to develop these sorts of finance.

Our evidence also shows that venture capitalists do invest smaller sums, reflecting both a willingness to invest in promising early seed stage ventures and to provide further follow-on financing when certain milestones have been reached. Based on our analysis, we offer the following recommendations.

6.1. University technology transfer offices and academic entrepreneurs

Resolving the problem of access to finance identified by technology transfer offices, and the difficulties in investing in USOs reported by venture capitalists, highlights the need to consider the role of university technology transfer processes. The availability of a stock of technology to be exploited, external expenditure on intellectual property protection, an experienced technology transfer team and business development capabilities may be particularly important.

USOs are heterogeneous. In those sectors where patenting is crucial, such as biotech and microelectronics, technology transfer offices need to ensure that intellectual property is clean, well-defined and properly protected before trying to raise external equity finance. A portfolio of patents may be needed to build a company, which includes both procedural, application and product claims, usually spread over different patents. If a claim is missing, it should be in-licensed or investigated to assess the extent to which the missing part belongs to the public domain. Where internal skills in this area are inadequate, adequate budgets may need to be provided to finance the cost of external advice.

Technology transfer offices should also not overemphasise the necessity for IP. Reflecting the heterogeneity of USOs, in some sectors, such as software development and services, IP is much less of an issue. While the ability to scale-up a prototype is key, few technology transfer offices are able to support this.

A separate but related issue concerns ownership and control in IP. Our evidence suggests a mismatch between the expectations of university technology transfer offices and external funders. *Prima facie* this is not surprising. The implications of this mismatch of expectations have not been fully recognised, which has major implications for external equity funded spin-outs. External equity providers are likely to be interested in generating financial returns within a specific time period, although business angels may be less exit oriented and more longer-term involvement focused than venture capital firms. Where TTOs see the potential return from a USO over a much longer time period, and include the potential contract research and the visibility of this for potential students, it may be more difficult to attract external equity.

In both the UK and continental Europe, universities and TTOs may have multiple objectives for their technology transfer through USOs, with implications for the creation of heterogeneous types of USOs with different external financial needs. To reconcile the mismatch of expectations with external finance providers, universities may need to consider more explicitly which USOs are targeted to generate financial returns and which for more social returns and design separate implementation strategies for each. These approaches may involve attempts to attract different types of external equity providers. USOs with high growth prospects in international markets may require investments from large VC firms with international networks and with high levels of technological expertise. Smaller local VC firms may not be able to exploit these opportunities. USOs where opportunities relate to more localised markets and/or with opportunities to create employment or modest returns may be more suited to smaller VC firms with public sector investment.

It is also important to recognise that not all inventions may be suitable for creation as a USO. Shane (2004), Dahlstrand (1999) and Mustar (1997) emphasise the importance of biotechnology and software sectors but it is also clear that USOs occur across a wide range of sectors. Shane (2004) notes that rather than the sector per se, the effectiveness of patents, the importance of complementary assets, the age of the industry, the degree of market segmentation and average firm size in an industry affect the likelihood of USO formation. Our interview evidence from TTOs indicates that the choice as to whether to license an invention or create a USO was principally related to which option generated the best returns, analysis of competitors and the market and the willingness of the academic entrepreneur to exploit the invention. A USO might be created when licensing to or a joint venture with a corporation was not feasible. While some VCs may have a policy not to invest in certain sectors, our findings from VC firms also emphasise overall quality of proposal, a clear route to market, market size, development stage and availability of a prototype as influences on investment decisions.

As indicated at the beginning of this paper, the problems and issues surrounding successful technology transfer are unsurprising given the relative dearth of experience in this area in many universities and in many financial providers. In attempting to overcome this skills gap, it is difficult for universities to ascertain the appropriate level and type of investment required. The return on investment in technology transfer capabilities is highly uncertain and the labour market for skilled technology transfer professionals is thin. However, our interviews reveal initiatives such as the scheme involving the MBA programme at the Swedish Chalmers University which suggest that, despite the many difficulties that still exist, the social return on these investments tends to be positive (see Jacobs et al., 2003).

Since part of the problem arises as a result of cultural differences between academia and private sector businesses specialising in commercialisation and innovation, the attributes and experience required are quite rare (Siegel et al., 2003b). Providing academics with experience of business and commerce, or transplanting private sector experience into an academic environment, may involve significant time lags before returns begin to emerge in terms of successful technology transfer. Venture capitalists agree that conditions are improving in universities but that there is also significant room for further improvement.

The apparent lack of investor readiness of USOs compared to other early stage high-tech ventures suggests important differences in the backgrounds of the entrepreneurs and their advisors. These differences lend support to the notion of recruiting more technology transfer officer skills from the private sector and/or to attract business developers from the private sector that want to step into the spin-out during the pre seed phase and develop the business plan. This raises implications for how universities should attract and retain technology transfer officers. A particular issue is that, if the valuation of the seed-stage venture is boosted through any input of patents, it might be difficult to involve business entrepreneurs who might end up with a very tiny equity share in this early phase. In our European interviews, we found similar problems in the technology transfer offices. In Finland, an efficient system was created, called DIILI, to attract business developers, which further elaborated the business plan and invested up to € 25,000 in the spin-out at start-up. They are not coaches as the technology transfer officers are, but they enter the company as CEO or COO in a very early phase and form a team with the researcher(s). They are coached closely by a team of consultants with backgrounds in diverse technologies and management disciplines. If universities are constrained to remunerate technology transfer officers in line with other elements of university bureaucracies, they may be unable to attract staff with the more entrepreneurial capabilities required to stimulate and develop successful spin-outs.

To enhance the attractiveness of universities to venture capital firms, there is a need to improve decision-making processes regarding venture capital investment in spin-outs. Universities may need to develop standard procedures and processes for creating spin-outs in order to speed up decision-making and offer greater transparency in the decision-making process. Universities also need to address the difficulties faced by VC firms in identifying key decision makers and having to deal with multiple stakeholders. This may be a function of the clarity of development of university's objectives and strategies towards USOs. Some universities may adopt a very conservative approach because of their concerns about accountability for public funds but universities more active in spinning-out companies adopt an approach to decision-making that is more in line with commercial business practice.

Relatively few universities have developed social capital in terms of ties with VC firms. The importance of these ties was stressed in the US context by Shane (2004). There is a need for further investment on the part of universities in developing these ties. A central aspect of the development of these relationships concerns the ability of universities to access the networks of venture capital firms. While academic entrepreneurs may have strong scientific networks, these may contain considerable redundant ties, for example in terms of research links, yet insufficient of the non-redundant contacts, such as finance and commercial skills, required for successful commercialisation (Burt, 1992). Difficulties in communicating between the networks of academic entrepreneurs and those of venture capitalists may arise because of differences in knowledge, codes, goals and assumptions. These problems create imperfections in the distribution of information, or structural holes, between networks. There is a need to bridge these structural holes by identifying individuals with the appropriate skills. Keil (2004) notes in the context of building corporate venturing capabilities that structural holes can be bridged by hiring people from venture capital and start-up communities. There is heterogeneity between universities in the distribution of these skills and the recognition of the need to access them.

Universities may find it more effective to develop links with private sector companies and leading entrepreneurs who have made successful investments in technology sectors wherever they are located in order to gain a better understanding of how to develop scientific research into commercial applications. These links may also help universities in deciding whether an invention is most appropriately developed as a license or a joint venture with a private sector corporation rather than as an independent spin-out with venture capital backing. Again, there may be a need to recruit people with the skills to make the link between the university and the commercial environment.

6.2. Venture capital firms and other finance providers

Our findings indicate that VC firms need to put greater emphasis on ensuring that universities understand their requirements. There may be benefits for venture capital firms to develop relationships with universities that are likely to generate a flow of possible investment opportunities. At present, few venture capital firms are developing links with more than one university. A longer term perspective may be required that involves developing potential deals from the seed stage acknowledging the possible need to make follow-on investments.

The heterogeneity of VC firms' human capital skills and financial resources affects investment behaviour. Our evidence of a greater likelihood of those VCs with higher public sector funding to reject ventures where there were concerns about the overall quality of the proposal and of the management team, suggests a more cautious approach that may be related to their weaker human capital and networks to recruit new team members. The emphasis on financial expertise and the limited experience in science and technology may have a detrimental influence on the type of early stage investment opportunity that are likely to be supported. There is a need to develop expertise in understanding the complexities of universities and appreciating how these differ from the traditional commercial environment in which external equity providers typically operate. The importance of brokering to identify executives, and customers and suppliers for the USO also emphasises the need for VC firms to have developed appropriate social capital to enable them to add value.

VC firms may also have diverse objectives that influence the kind of USOs in which they will invest. VCs appear to have similar approaches to rejecting proposals and attitudes to risk and reward but those with greater public sector funding seem more prepared to invest in earlier stage ventures but are more concerned about the skills of the entrepreneurs. Our European survey suggests that larger VC funds tend to invest in later stages and not in the early or seed phase, on the average. This means that many spin-outs are not able to go to the next rounds with the same investor.

6.3. Government policies

In the UK, public–private partnerships such as UCFs provide incentives for investors to increase their exposure to spin-outs and hence to learn to engage with universities. These funds have a continuing role in focusing on achieving proof of concept for technologies and enhancing the ability of spin-outs to reach investor readiness sufficient to access funding from early stage investors. Further development of this kind of funding could facilitate involvement in the market by more VCs but may require the development of close relationships between universities and venture capitalists. Benefits may also be achieved from enabling these funds to reduce the complexity of negotiating with existing stakeholders when spin-outs reach the stage where they require larger amounts of venture capital funding.

In other countries such as Sweden and Finland, government initiatives provide both pre-seed capital and networking and coaching for high-tech start-ups and spin-outs. This is a step beyond the general ambition of the UCFs although some individual UCF implementations tend to provide quite extensive coaching. It is questionable whether public–private seed capital money can be used to pay for coaching expenses. In other countries, the investment managers are not paid by the university fund but by public schemes such as the National Incubation Programme in Sweden. Such a scheme does not only cover the more classic investment management activities such as deal flow generation and selection. A large part of the money is used for the salaries of individual coaches that guide the development of business plans for pre-revenue USOs.

In addition, the VC literature shows that having complete and complementary teams is a very important condition before investments are made (Muzyka et al., 1996). Again, there is a market distortion here, because VCs do not want to take the risk of investing in putting these teams together. TTOs observe that attracting commercial management and/or business angels – which play an active role in the company – is one of the key problems they face in creating USOs. However, the university venture capital funds do not have sufficient investment management capacity to provide a sufficient amount of networking. Again, it is questionable whether this problem can be tackled by public–private partnerships. The DIILI matching programme in Finland, is developed specifically to look for business developers and adequate business angels. This DIILI programme is subsidised in the context of market failing to optimise networking.

The direct involvement of government in the pre-seed capital stage, possibly complemented with coaching and networking support is part of a push strategy to start-up spin-outs with an optimal configuration. In addition, fiscal measures might be very promising in the light of pull activities to encourage the business angel community to increase their interest in spin-outs. In line with the Dutch "Tante Agaath" initiative, a yearly income tax deduction could be proposed for individuals who invest up to € 25,000 in a pre-revenue start-up. The Dutch Scheme is so successful that the government had to put a limit on the number of applications per year.

The above measures assume that universities and public research organisations provide a fertile ground for spin-outs. This is not to be taken for granted. If these institutes are supposed to generate spin-outs on a more systematic basis, a further key issue concerns the need to develop the skills and the culture required to promote academic entrepreneurship and attract VC finance. Government intervention has generated various initiatives to promote entrepreneurship education for both students and staff. In the UK, for example, entrepreneurship classes are being encouraged in masters, under-graduate and executive teaching. Some students want to start businesses and some schemes, such as the TOP Scheme at the University of Twente in The Netherlands, provide direct support for such efforts. In general, though, there appears to be a greater need for the integration of entrepreneurship in students curricula. Initiatives have often involved the provision of relatively short term funding for what is essentially a long term process. The design of permanent third stream funding to universities may be critical in determining the efficient engagement of universities with knowledge transfer and, in particular, technology transfer.

Notwithstanding the above developments, there remain concerns that the goal of achieving substantial revenues from the commercialisation of university research has yet to be achieved and questions over whether commercialisation of university inventions in this way is appropriate (e.g. Nelson, 2001). Although discussion of this issue is outside the scope of this paper, there is a need for monitoring the contribution of commercialisation activities in both the short and long run and of the contribution of USOs in particular. This paper has suggested policies to strengthen the role of both TTOs and venture capital firms. It has also indicated a need for realism concerning the extent of USOs that are likely to generate high growth and high returns. In this context, it may be important to consider the wider social returns from USOs and other forms of university commercialisation rather than just direct financial returns. Shane (2004), for example, quotes evidence on both the direct and indirect benefits of USOs.

6.4. Institutional variations between countries

Our evidence has shown some consistency across countries in the problems faced by USOs that may be expected to seek venture capital in actually obtaining it. The suitability of USOs for venture capital financing and the type of venture capital financier that is appropriate may relate more to the scientific base of the university and to the heterogeneous nature of USOs than to the country context per se (Clarysse et al., 2005).

One of the largest institutional differences between countries is the presence of a Bayh-Dole kind of Act. In Germany for instance, this kind of Act was only passed in 2003. Before that, universities had little incentives to set up an interface service responsible for the management of its IP. In those countries where a national version of the Bayh-Dole Act was voted, a professionalisation of the interface service was the result. In this professionalisation wave, IP was typically given to spin-outs but had to be valued (e.g. as a percentage of equity). This is usually only possible if capital can be invested in spin-outs and many spin-outs started to look for different forms of external equity. Universities were assisting in this and tried to create different support schemes, with or without the help of the government. In different countries, these initiatives are institutionalized in a very different form. A recent expert group in preparation of the 3% norm at the European Commission has illustrated the importance of this kind of Venture Capital to finance technology transfer and R&D (EC, 2003).

Industry or technology specific differences between countries seem to play less of a role, although they might do at an organisational level. But even within specific technological domains, there is a difference between types of spin-outs. In technologies ranging from biopharma to industrial software development, capital intensive spin-outs as well as start-ups that only need very small amounts of capital are found. To our knowledge, there is no scientific evidence that the industry variable has explanatory power in making differences between spin-out types. The existing industry specific studies depart usually from the assumption that there *is* a difference, but they do not validate this assumption. In the European sample we find that software spin-outs made up about 50% of the spin-out population in the period 1995–2000. Differences are minor between countries ranging from a mere 40% in Germany towards almost 60% in Sweden. Biotech and Medical related technologies count for 10–20%. Micro-electronics are also 10–20% and the rest is a heterogeneous set of others.

6.5. Research implications

Our work adds to existing research regarding the spin-out process and suggests a number of avenues for further research. Much research attention has focused on the complexities of the process of identifying IP, proof of concept, markets and entrepreneurial teams. A key determinant of the growth and value generation associated with spin-outs concerns access to finance. In a context of highly restricted personal funds and funds provided by parent universities, access to venture capital assumes major importance. Yet, our evidence suggests that, on the one hand, TTOs and spin-outs need to complement existing activities to do more to develop processes, skills and understanding of what is required to attract venture capital. On the other hand, informational asymmetries and differences between the university and private sector high-tech contexts have contributed to a limited flow of venture capital to spin-outs.

Adopting a financial pecking order hypothesis perspective, we find a mismatch between the demand and supply side of the spin-out market. VC firms appear to be more likely to fund spin-outs once they have become established and this suggests that further research is warranted that examines the complementarities and substitutabilities between VC and other forms of financing. Part of the analysis might focus on the adoption of other business models in spin-outs to generate enough revenues to bridge the equity gap or to develop other financing mechanisms that increase internal and debt funding. As aforementioned, initiatives have been taken in Sweden and Finland to facilitate these financing forms. However, more research is needed to analyse their effectiveness. Also, part of the analysis of complementarities may need to consider the extent and nature of syndication between different types of finance provider and how this changes over the development of the spin-outs. For example, what problems arise in syndicating between public and private sector funds providers who may have different return and time horizon objectives?

Our discussion has also alluded to the problems in developing relationships and understanding between universities and external equity providers. This adds further insights into the complexities of the commercialisation process. It suggests an important link between social capital, human capital and financial capital in this process. There is, therefore, a need to develop research that examines the nature of boundary spanners between the university and external finance providers. Who fulfils this role and what human capital do they possess? What are the barriers to the fulfilment of this role?

Different types of universities may create different types of spin-out. Similarly, some universities may create different types of spin-outs with different objectives and growth prospects that have different financial requirements. Further research is needed that considers systematically the process of matching the appropriate type of external finance provider with particular types of spin-outs at different phases in their development. Again, this research needs to consider the nature of the networks of universities and external finance providers. This research needs to examine in depth differences both within and between countries, including the implications of different types of research regimes between countries on the types of opportunities for USOs that are being generated.

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